HVF LEVEL-5 SOVEREIGN C2 OPERATING CONSOLE (PROJECT EBONY)
DoD CDAO Tradewinds Solutions Marketplace // Turnkey Tactical & Industrial Operating System Dossier
Prime Contractor: Humphrey Virtual Farms LLC | **CAGE:** ‘1AHA8‘ | **UEI:** ‘S1M4ENLHTDH5‘
CEO & Apex Systems Architect: Jeffery Humphrey
Headquarters: Madill, Oklahoma 73446 | **Contact:** ‘[REDACTED_PUBLIC_RELEASE_CONTACT]‘

Technology Readiness Level: **TRL 7** (Operational System Prototype Verified on Bare-Metal Enclave)
Associated Tradewinds Dockets: ‘10-26-3952‘ (‘Status: Received‘) & ‘9-26-3703‘
Master C2 Merkle Root (SHA-256): ‘cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e9150949a‘
Timestamp: ‘2026-10-25 19:16:36 UTC‘
Statutory & Data Rights Governance: DFARS ‘252.227-7018‘ / ‘252.227-7014(a)(15)‘ / ‘227.7202‘, DFARS ‘252.204-7012‘, NIST SP ‘800-171‘, OSHA ‘1910.212‘, FAA ‘14 CFR Part 107‘, Oklahoma Title 61

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1. Executive Tradewinds Submission Abstract & TRL 7 Host Verification Notice

Title: ‘Project Ebony: Level-5 Air-Gapped Sovereign C2 Operating Console for Deterministic AI, 36.75ns Robotics, UAS Diagnostics, and SCADA Security‘

Important Procurement & Licensing Notice for Tradewinds Evaluators and Contracting Officers:
The **HVF Level-5 Sovereign C2 Operating Console (Project Ebony)** documented in this dossier serves as the verified **TRL 7 Operational Host Environment** demonstrating that Humphrey Virtual Farms LLC’s cognitive routing and physical safety architectures (‘Tradewinds Dockets 10-26-3952 & 9-26-3703‘) are fully operational on bare-metal hardware across **631 compiled Python/C modules**, **26 specialized Sidebar Command Modules**, **72 verified operator controls**, and **83 passing unit test suites (‘100.0% parity‘)**.

* **Base Pilot Entry Scope (‘Docket 10-26-3952‘):** The entry pilot price submitted on Tradewinds Docket ‘10-26-3952‘ applies **strictly and exclusively to ‘CLIN 0001‘ (‘Module #01: Sovereign Command Starter Pilot‘ on ‘1‘ Single Bare-Metal Node)**.
* **Unbundled Modular Licensing (‘CLIN 0002‘ - ‘CLIN 0026‘):** Each of the remaining **25 Command Modules (‘Modules #02 through #26‘)** is a separately priced, cryptographically gated software SKU developed exclusively at private expense under **DFARS 252.227-7018** and **DFARS 227.7202**. Government and commercial customers may procure individual modules a la carte **per node** or procure pre-packaged Domain / Enterprise Bundles (‘CLIN 1001‘ - ‘CLIN 1004‘) via sole-source Other Transaction Authority (10 U.S.C. 4022) or FAR Part 12 commercial contract modifications.

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2. Strict Per-Node / Per-Chassis / Per-Seat Hardware-Locked Licensing Mandate

Pursuant to **DFARS 252.227-7018**, **DFARS 252.227-7014(a)(15)**, and **DFARS 227.7202**, all software modules within the HVF Sovereign C2 Operating Console are licensed on a **strictly per-node, per-chassis, or per-seat basis**:

1. **One License = One Physical Machine (‘1 Node‘):** A single Node License authorizes execution on **one (1) physical bare-metal computer, server, robotic controller, or Tactical Operations Center (TOC) workstation** bound to a unique hardware UUID and SQLite WORM ledger (‘matrix_ledger.db‘). Copying, cloning, virtualizing across multiple hosts, or distributing a single-node license across multiple computers, buildings, or military installations without procuring additional Node Licenses is strictly prohibited.
2. **Robotic, UAS, & Automotive Hardware Nodes:** ‘CLIN 0006‘ (‘Robotics C2‘), ‘CLIN 0007‘ (‘Drone Diagnostics‘), and ‘CLIN 0008‘ (‘Auto CAN-Bus C2‘) are licensed **per physical robot controller/UGV chassis**, **per UAS Ground Control Station (GCS)**, and **per maintenance bay node**, respectively.

| **CLIN 0008** | **Mod #08:** ‘Sovereign Auto & CAN-Bus C2 (SAE J1939)‘ | Per Fleet Maint
enance Bay / Diagnostic Node | **$85,000 / yr** | **$7,600 / mo** |
| **CLIN 0009** | **Mod #09:** ‘Grain Silo Aeration & 480V Contactor C2‘ | Per Silo Complex
/ 480V MCC PLC Node | **$75,000 / yr** | **$6,800 / mo** |
| **CLIN 0010** | **Mod #10:** ‘Omni-Industry Matrix & Title 61 UAS Deliverables‘ | Per Engi
neering District / Inspection Node | **$95,000 / yr** | **$8,900 / mo** |
| **CLIN 0011** | **Mod #11:** ‘Sovereign Field Academy (1,420 Verified Exam Qs)‘ | Per 50
-Seat Training Cohort *(or $295/mo per Seat)* | **$125,000 / yr** | **$11,200 / mo** |
| **CLIN 0012** | **Mod #12:** ‘HVF Master Auto & Small-Engine Academy Deck‘ | Per Technical
Training Lab Node (Up to 25 Seats) | **$65,000 / yr** | **$5,900 / mo** |
| **CLIN 0013** | **Mod #13:** ‘Sovereign Dispatch Deck (Air-Gap IMAP/SMTP)‘ | Per Secure
Communications Enclave Node | **$55,000 / yr** | **$4,950 / mo** |
| **CLIN 0014** | **Mod #14:** ‘Sovereign Comms Deck (Dual-Core Audio/Triage)‘ | Per Tacti
cal TOC Comms Node | **$50,000 / yr** | **$4,500 / mo** |
| **CLIN 0015** | **Mod #15:** ‘Media Matrix & Predict-and-Act Autonomous Loop‘ | Per Autono
mous Operations Node | **$65,000 / yr** | **$5,900 / mo** |
| **CLIN 0016** | **Mod #16:** ‘Sovereign Asset Synthesis Engine‘ | Per Local Render Worksta
tion Node | **$40,000 / yr** | **$3,600 / mo** |
| **CLIN 0017** | **Mod #17:** ‘LinkedIn Executive Broadcast Engine‘ | Per Executive Communi
cations Node | **$30,000 / yr** | **$2,750 / mo** |
| **CLIN 0018** | **Mod #18:** ‘Python Execution Sandbox & Quarantine Harness‘ | Per Develop
er / Audit Node | **$45,000 / yr** | **$4,100 / mo** |
| **CLIN 0019** | **Mod #19:** ‘Empire Config & Pilot Governance Vault‘ | Per Administrative
Root Node | **$35,000 / yr** | **$3,200 / mo** |
| **CLIN 0020** | **Mod #20:** ‘Level-5 Ext: Action Desk Tactical Dispatcher‘ | Per Sector C
ommand Node | **$30,000 / yr** | **$2,750 / mo** |
| **CLIN 0021** | **Mod #21:** ‘Level-5 Ext: NOAA Weather Radar & EMC Sync‘ | Per Agricultur
al / Base Meteorology Node | **$40,000 / yr** | **$3,600 / mo** |
| **CLIN 0022** | **Mod #22:** ‘Level-5 Ext: ISO 32000-1 WORM PDF Compiler‘ | Per Compliance
Documentation Node | **$35,000 / yr** | **$3,200 / mo** |
| **CLIN 0023** | **Mod #23:** ‘Level-5 Ext: Autonomous Lead Scheduler‘ | Per Commercial Out
reach Node | **$25,000 / yr** | **$2,250 / mo** |
| **CLIN 0024** | **Mod #24:** ‘Level-5 Ext: Chronos Companion Telemetry‘ | Per Watch-Floor
Operator Node | **$30,000 / yr** | **$2,750 / mo** |
| **CLIN 0025** | **Mod #25:** ‘Level-5 Ext: 48-Plenum Silo PLC Synchronizer‘ | Per Industri
al Facility PLC Node | **$35,000 / yr** | **$3,200 / mo** |
| **CLIN 0026** | **Mod #26:** ‘Level-5 Ext: Cryptographic Merkle Verifier‘ | Per Independen
t Audit / Inspector General Node | **$20,000 / yr** | **$1,850 / mo** |
| **TOTAL** | **All 26 Modules Purchased Individually (Unbundled)** | **1 Single Complete
Enclave Node (All 26 CLINs)** | **$1,630,000 / yr** | **$148,950 / mo** |

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4. Pre-Packaged Domain & Master Enterprise Bundles (‘CLIN 1001‘ - ‘CLIN 1004‘)

| Bundle CLIN | Official Bundle Name & Included Modules | Hardware Node Scope | Annual Gov /

Enterprise Price | Monthly Commercial Lease |
| :--- | :--- | :--- | :--- | :--- |
| **CLIN 1001** | **Tactical Robotics, UAS, & CAN-Bus Autonomy Bundle**
(Includes Modules ‘#01, #06, #07, #08, #10‘) | **1 Single Robotics / Autonomy Command Node** | **$385,000 / yr per Node**
(Saves $79,500/yr) | **$34,500 / mo** |
| **CLIN 1002** | **Critical Infrastructure, 60Hz Microgrid, & 480V SCADA Bundle**
(Includes Modules ‘#01, #03, #09, #20, #21, #25‘) | **1 Single Substation / Facility SCADA Node** | **$295,000 / yr per Node**
(Saves $79,500/yr) | **$26,500 / mo** |
| **CLIN 1003** | **Accredited Workforce & Technical Readiness Academy Bundle**
(Includes Modules ‘#11, #12, #22‘ + All 4 Textbooks) | **1 Training Server Node (Up to 100 Named Student Seats)** | **$195,000 / yr per Cohort** | **$17,500 / mo** |
| **CLIN 1004-A** | **Master Turnkey Sovereign C2 Operating Console (Single-Node All-Access)**
(Unlocks All ‘26‘ Modules & All ‘72‘ Verified Controls) | **1 Single Bare-Metal TOC / Command Enclave Node** | **$895,000 / yr per Node**
(Saves $735,000/yr vs Unbundled) | **$79,500 / mo** |
| **CLIN 1004-B** | **Military Installation / Command-Wide Fleet Enterprise License**
(Unlocks All ‘26‘ Modules Across Up to ‘25‘ Hardware Nodes at 1 Base) | **Up to 25 Enclave / Robotic / SCADA Nodes (1 Installation)** | **$2,450,000 / yr per Base** | **$215,000 / mo** |

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5. 5-Minute Tradewinds Video Walkthrough Script (For CDAO Evaluator Review)

* **0:00 - 0:45 | Executive Identification & Cryptographic Seal (‘verify_sovereign_seal.py‘ & ‘run_all_83_tests.py‘):**
  Show Windows PowerShell executing ‘verify_sovereign_seal.py‘ (‘631 / 631‘ modules, Merkle Root ‘cc1ee1c8...949a‘) and ‘run_all_83_tests.py‘ (‘83 / 83 PASSED‘). State Entity (‘Humphrey Virtual Farms LLC‘, ‘CAGE: 1AHA8‘, ‘UEI: S1M4ENLHTDH5‘), TRL 7 operational status, and that ‘Docket 10-26-3952‘ provides ‘CLIN 0001‘ entry access into the 26-module console.
* **0:45 - 1:45 | Master C2 Cockpit & Zero-Hallucination Live WAN Command Stream (‘CLIN 0001‘ & ‘CLIN 0003‘):**
  Show the live browser console on Port ‘8501‘. Click ‘System Readiness‘ and ‘Swarm Triad Audit‘, then run a live query to demonstrate real-time HTTPS wire grounding and the ‘VERIFIED LIVE WAN WIRE TELEMETRY‘ citation block.
* **1:45 - 3:00 | Top-Level ‘Sovereign Robotics & Kinematics C2‘ (‘CLIN 0006‘ // ‘EBONY-RTOS‘ 36.75ns Guillotine):**
  Click ‘Sovereign Robotics & Kinematics C2‘ in the sidebar. Switch across the 6-Axis Industrial Arm, 28-DOF Bipedal Humanoid, and Defense/Ag Tracked UGV. Trigger ‘Execute Emergency 36.75ns Crowbar Trip Test‘ (‘0.0551 um‘ stopping distance vs ‘225.0 mm‘ PLC) and show the new SHA-256 row in Tab 4 (‘WORM Robotics Telemetry Ledger‘).
* **3:00 - 4:00 | Autonomous UAS Sortie & Civil Infrastructure Certification (‘CLIN 0010‘):**
  Select docket ‘HVF-CIV-ROA-404793C1‘, click ‘Dispatch Drone Sortie‘ (‘HVF-SPECTRE-X8‘), and click ‘Synthesize Deliverable‘ to generate the SHA-256 sealed Title 61 engineering rep

ort (‘87.0 / 100 Sufficiency‘).
* **4:00 - 5:00 | CEO Authorization Airlock (‘CLIN 0005‘) & Accredited 1,420-Question Academy (‘CLIN 0011‘):**
Show ‘CEO Authorization Gate‘ (‘698‘ modules inspected, ‘0‘ defects) and ‘Sovereign Field Academy‘, concluding with immediate readiness for per-node and enterprise OTA contract awards under Tradewinds.

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6. 5-Year Commercial Viability, Multi-Node ARR Trajectory (FY 2027 - FY 2031), & DoD Zero-Cloud Cost Avoidance

Pursuant to FAR Part 12 Commercial Item price reasonableness and DoD CDAO Tradewinds Commercial Viability criteria, Humphrey Virtual Farms LLC operates a high-margin (‘>92%‘ gross software margin, ‘$0‘ third-party cloud hosting drain) recurring per-node licensing model across three 5-year deployment trajectories (1 Base Year + 4 Option Years):

5-Year Deployment Tier	Active Installed Base by Year 5 (Per-Node / Per-Chassis / Per-Cohort)	Year 1 ARR	Year 3 ARR	Year 5 ARR	5-Year Cumulative Revenue
Tier 1: Conservative (Regional & State DOT / Guard Scale)	‘12‘ Base Pilots (‘CLIN 0001‘), ‘10‘ SCADA/UAS Nodes, ‘6‘ Robotics Nodes (‘CLIN 0006‘), ‘6‘ Academy Cohorts, ‘3‘ Full Consoles (‘CLIN 1004-A‘)	**$559,000**	**$2,899,000**	**$5,689,000**	**$14,990,000**
Tier 2: Baseline Target (Tradewinds Multi-Base & Depot Scale)	‘25‘ Base Pilots, ‘30‘ SCADA/UAS Nodes, ‘25‘ Robotics Chassis Nodes, ‘15‘ Academy Cohorts, ‘8‘ Full Consoles, ‘3‘ Base 25-Node Fleet Licenses (‘CLIN 1004-B‘)	**$1,690,000**	**$10,250,000**	**$24,172,500**	**$54,850,000**
Tier 3: Aggressive (DoD Command-Wide & National Fleet Scale)	‘50‘ Base Pilots, ‘80‘ SCADA/UAS Nodes, ‘90‘ Robotics/UGV Chassis Nodes, ‘40‘ Academy Cohorts, ‘18‘ Full Consoles, ‘10‘ Base 25-Node Fleet Licenses	**$4,590,000**	**$28,900,000**	**$70,285,000**	**$153,200,000**

* **FAR Part 13 Simplified Acquisition Alignment:** Every individual module (‘CLIN 0001‘ through ‘CLIN 0026‘, ‘$20,000/yr‘ to ‘$185,000/yr‘ per node) is priced strictly below the **‘$250,000‘ Simplified Acquisition Threshold**, enabling rapid squadron-, depot-, and district-level procurement without multi-quarter formal source-selection delays.
* **Zero-Cloud DoD Lifecycle Cost Avoidance (‘>42.5%‘ Savings):** Because Project Ebony executes 100% on air-gapped bare-metal hardware with zero recurring commercial cloud egress/compute surcharges (no AWS GovCloud or Azure Government meter), procuring agencies save an estimated **42.5% to 58.0%** in 5-year Total Ownership Cost (TCO) compared to cloud-tethered C2 platforms.

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7. Day-1 Mobilization Invoicing, Payable Milestone Schedule (10 U.S.C. 4022 / FAR 52.212-4), & 15-Day Accelerated Payment (FAR 52.232-40)

To ensure immediate hardware provisioning and enclave mobilization upon contract award without vendor out-of-pocket pre-financing, all awards under ‘Docket 10-26-3952‘ and ‘CLIN 0001‘ - ‘CLIN 1004‘ execute under the following federal commercial payment terms:

1. **Simplified Acquisition & Single-Node COTS Licenses (‘<= $250,000‘):** Invoiced **100% on Day 1 of Contract Award** upon electronic delivery of the WORM SHA-256 Hardware-Locked License Certificate (‘FAR 52.212-4‘), payable via Government Purchase Card (‘FAR 13.301‘, 24-48 hr settlement) or PIEE/WAWF with mandatory **15-Day Accelerated Small Business Payment (‘FAR 52.232-40‘)**.
2. **Multi-Node & Enterprise Console Awards (‘> $250,000‘ under 10 U.S.C. 4022 OTA / FAR Part 12):**
   * **Payable Milestone #01 (Day 1 of Award ? ‘50%‘ of Total Award):** Electronic issuance of Master Cryptographic License Certificate, WORM Baseline Attestation, and Bare-Metal Enclave Hardware Mobilization.
   * **Payable Milestone #02 (Day 30 ? ‘30%‘ of Total Award):** Delivery of pre-configured bare-metal enclave hardware and site-specific TRL 8 interface binding.
   * **Payable Milestone #03 (Day 60 ? ‘20%‘ of Total Award):** Final operational verification and Field Academy cohort activation.

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8. Post-Award TRL 8/9 Funded Engineering & Site Integration Roadmap (Day 1 to Day 90)

Project Ebony’s software enclave is 100% complete and frozen at **TRL 7** (‘631 / 631‘ modules, ‘83 / 83‘ unit test suites). Following contract award and Day-1 mobilization payment settlement (‘FAR 52.232-40‘ / ‘10 U.S.C. 4022‘), Humphrey Virtual Farms LLC executes six customer-funded **TRL 8 (Site HIL Integration)** and **TRL 9 (Full Operational Accreditation)** work packages out of the **18% Setup Hardware** and **15% TRL 8/9 Integration** budget allocations:

* **‘WP-1‘ (Days 1-15 | TRL 8):** Customer bare-metal appliance provisioning, NVMe WORM array formatting, and Hardware UUID cryptographic license binding.
* **‘WP-2‘ (Days 10-30 | TRL 8):** Physical SiC MOSFET ‘36.75ns‘ hardware guillotine relay board assembly and robotic actuator bench calibration (‘CLIN 0006‘).
* **‘WP-3‘ (Days 15-45 | TRL 8):** Site-specific Modbus-TCP/DNP3 60Hz microgrid relay (‘CLIN 0003‘), 480V contactor (‘CLIN 0009‘), and SAE J1939 CAN-Bus (‘CLIN 0008‘) register mapping.
* **‘WP-4‘ (Days 20-45 | TRL 8):** Local UAS optical payload ingestion (‘CLIN 0007‘) and Oklahoma Title 61 / base civil engineering GIS boundary binding (‘CLIN 0010‘).
* **‘WP-5‘ (Days 30-60 | TRL 8/9):** Command-specific watch-floor role customization and automated ISO 32000-1 ‘%PDF-1.4‘ shift-ledger formatting (‘CLIN 0022‘).

* **‘WP-6‘ (Days 45-90 | TRL 9):** Site Authority to Operate (ATO / NIST SP 800-53 RMF) package sign-off and 50-seat Field Academy cohort deployment (‘CLIN 0011‘).